

Quizzes

The ultimate tensile strength of a polymer is most strongly influenced by:

- a) M_w .
- ☒ b) M_n .
- c) M_z .
- d) M_e .

Quizzes

The viscosity of a polymer melt is most strongly influenced by:

- ☒ a) M_w .
- b) M_n .
- c) M_z .
- d) M_e .

Quizzes

The Young's Modulus of an amorphous polymer in the rubber plateau, is most strongly influenced by:

- a) M_w .
- ☒ b) M_n .
- c) M_z .
- d) M_e .

Quizzes

The melt elasticity of a polymer melt is most strongly influenced by:

- a) M_w .
- b) M_n .
- ☒ c) M_z .
- d) M_e .

Quizzes

Which of the following side group arrangements can lead to polymers that cannot crystallise?

- a) Isotactic.
- ☒ b) Atactic.
- c) Syndiotactic.
- d) Symmetric.

Quizzes

Polyethylene (PE) is more sensitive to creep at room temperature than polycarbonate (PS) because:

- a) PE is semi-crystalline.
- b) PE is amorphous.
- c) PE has a higher T_g .
- ☒ d) PE has a lower T_g .

Quizzes

With increasing temperature, above the glass transition temperature, the modulus of a highly stretched cross-linked rubber will:

- a) Decrease substantially.
- b) Decrease slightly.
- ☒ c) Increase slightly.
- d) Remain constant.

Quizzes

Select the incorrect statement:

- a) Polystyrene (PS) is brittle because it crazes ✓
- b) Polycarbonate (PC) is tough because it can yield ✓
- c) Mechanical properties of PS and PC are dependent on testing temperature ✓
- ☒ d) Mechanical properties of PS and PC are independent on testing speed (strain rate)

Questions

Explain why Polystyrene (PS) disposable cups are transparent.

Amorphous

Questions

Explain why Polystyrene (PS) expanded foams are not transparent

Crystalline

Questions

Explain why poly(ethylene terephthalate) (PET) bottles are transparent.

Semi-crystalline

Questions

For each of the following applications, select a polymer and manufacturing process, briefly justifying your choice.

Milk Bottles

PE
A
 $> T_g$
injection
moulding



ABS

A

$< T_g$

injection moulding

Lego® bricks



Electrical Cable (e.g. CoAx)

silicon
rubber

C $> T_g$

co-extrusion



Electrical Sockets

Epoxy

A

$< T_g$

injection moulding



Questions

For each of the following applications, select a polymer and manufacturing process, briefly justifying your choice.

Rubber

A

$< T_g$

demoulding

Tyres



PET

A

$< T_g$

film extrusion

Plastic Bags



Fishing Line

Uhmwpe

A

$< T_g$

fibre spinning



Non-metallic loaf container

silicon rubber

C

$< T_g$

injection

moulding



Questions

For each of the following applications, select a polymer and manufacturing process, briefly justifying your choice.

PPTA aramid fibre spinning

Bullet-Proof Vest (external fabrics)



PP

Foams



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